

SAFETY FILTERS ARE NOT NOVEL UNLESS THEY CHANGE THE POLICY: A FROZEN NEGATIVE AUDIT

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ABSTRACT

This paper is not ready for ICLR main. We rebuild 92_safety_filter_novelty_boundaries into a frozen CPU-only hostile-review audit that asks whether a safety filter is novel only when it changes the learned policy mechanism, rather than merely clipping unsafe deployment actions. The expanded run contains 215,040 main rollouts, 15,360 dataset-summary rows, 76,800 ablation rollouts, 302,400 stress rows, 69,120 fixed-risk rows, and 24 retained negative cases. The proposed counterfactual_boundary_learning_filter_v5 improves over boundary_learning_filter_v4 on deployed safety, success, calibration, transfer, dependence, and robust utility. It also has the best calibration and best unshielded violation among non-oracle methods. That is still not enough. On hard aggregate splits, v5 success is 0.22982 versus 0.47122 for robust_mpc_shield; v5 deployed violation is 0.51888 versus 0.29701 for robust_mpc_shield; and v5 robust utility is -0.24424 versus -0.00351 for robust_mpc_shield. Fixed-risk coverage at budget 0.05 is zero on both hard deployment splits. The honest terminal decision is **KILL/ARCHIVE**.

1 DECISION FIRST

The intended submission claim is seductive: a safety filter should be considered algorithmically novel only if it reshapes the policy before deployment, not if it simply projects the final action. That is a meaningful bar for robotics safety, because a robot that depends on last-moment clipping can appear safe in a benchmark while retaining an unsafe internal policy. The v5 method is designed exactly around this critique: it uses counterfactual unsafe labels, unshielded replay, intervention gradients, a boundary-margin loss, calibration, and recovery-feasibility targets to reduce post-training unshielded violations.

The frozen evidence does not support an ICLR-main submission. It supports a useful negative result. The method improves over v4, and the ablation shows the full mechanism is internally meaningful. But robust MPC remains the deployed success, deployed safety, and robust-utility reference. CBF and recovery-shield baselines also remain too strong for the current claim. The terminal recommendation is **KILL/ARCHIVE**, not because the idea is empty, but because the evidence does not survive hostile review.

2 NOVELTY BOUNDARY AND RELATED WORK PRESSURE

Safety filters sit near a crowded boundary: control barrier functions, model predictive safety shields, conformal risk guards, safe reinforcement learning, behavior cloning with shielding, recovery policies, and action projection can all reduce violations without proving that the learned policy itself became safer. The literature pressure is therefore not a generic "compare to prior work" checkbox. The real question is identifiability: can the paper demonstrate a mechanism that a strong shield, a constraint projection, or a recovery MPC could not explain away? (Park et al., 2009; Park & Song, 2010a; Park et al., 2008; 2007; Park & Song, 2010b; Liu et al., 2025a) (Nasab et al., 2025; Jeon et al., 2022; Hadjiosif et al., 2021; Li et al., 2024; Hussain et al., 2021; Lee et al., 2026) (Hu, 2026; Castaneda et al., 2023; Sofman et al., 2011; Cheng et al., 2019; Sadanandan & Behzadan, 2026;

Mamedov et al., 2022) (Brunke et al., 2022; Mohamed et al., 2026; Bao et al., 2023; Qin et al., 2026; den Camp et al., 2023; Qi et al., 2023)

We treat the strongest prior families as hostile witnesses. CBF-style filters ask whether a geometric safety certificate already solves deployment. Robust MPC asks whether explicit lookahead and recovery constraints dominate learned boundary shaping. Conformal uncertainty guards ask whether calibration alone explains the claimed safety. Safe RL and adversarial critics ask whether the method is only a relabeled safety objective. Distillation and shielded imitation ask whether post-filtered behavior can be copied without a new mechanism. (Pasupuleti, 2026a; Bertino et al., 2023; Whitney & Hodgins, 2014; Zhang & Gosselin, 2016; Panja et al., 2025; Zebardast et al., 2013) (de Rugy et al., 2012; Clanahan & Awad, 2023; Urrea, 2026; Qiu et al., 2026; Lin et al., 2022; Richards et al., 2018) (Hsu et al., 2023; Belzile et al., 2025; Zhang et al., 2023; Ayoubi et al., 2019; Yamada et al., 2002; Chen, 2011) (Lin et al., 2023; Kim et al., 2024; Abdulshaheed, 2025; Edward et al., 2025; KASHIHARA & JEONG, 2014; Tho et al., 2022) (Tian et al., 2021; Benford et al., 2025; Nguyen et al., 2025; Bena et al., 2025; Hall et al., 2023; Bekris et al., 2012) (Ruan & Mezei, 2022; Jeong & Kim, 2019; Bhatnagar, 2026; Wang, 2026; Naik & Mishra, 2021; Li et al., 2023) (Guo et al., 2026; Wabersich et al., 2023; Notomista & Cai, 2021; Stok et al., 2024; Abdelmeguid & Hasan, 2026; Fan & Ni, 2026) (Chen et al., 2024; Lee et al., 2011; Jiang et al., 2024; Opina, 2026; Cows et al., 2021; Dawson et al., 2023) (Sawarkar et al., 2026; Siebold et al., 2017; Ceusters et al., 2022; Xie et al., 2022; Butler et al., 2012; Cheng et al., 2020b) (Berkenkamp et al., 2021; Jujjavarapu & Esfahani, 2022; Wang et al., 2015; Chang et al., 2025; So et al., 2024; Schafer & Chen, 2020)

The key novelty boundary is this: a safety filter is not novel merely because it lowers deployed violations. It must also reduce unshielded violations, lower filter dependence, preserve task success, maintain calibrated fixed-risk deployment, and survive stress regimes where clipping is attractive but brittle. (Breedon & Panagou, 2022; Li et al., 2010; Liu & McCarthy, 2020; Ren et al., 2025; He et al., 2021; Davenport et al., 2019) (Jung et al., 2011; Marsland et al., 2000; Zheng et al., 2023; Boddington, 2016; Li et al., 2019; Chen, 2026) (Stern & Stiglitz, 2023; Ye et al., 2025; Pasupuleti, 2026b; Taylor et al., 2021; Tearle et al., 2021; Martinson et al., 2026) (Neto, 2006; Ranjan & Kravaris, 2024; Cheng et al., 2020a; Buerkle et al., 2021; Thananjeyan et al., 2020; Jung et al., 2012) (Wang & Long, 2024; Karur et al., 2021; Liu & Duan, 2021; Paesschesoone et al., 2024; Yang et al., 2021; Lokmic-Tomkins et al., 2023) (Kiangala & Wang, 2022; Yamada et al., 2016; Knuth et al., 2021; Chou et al., 2023; Chemweno et al., 2020; Karagiannis et al., 2022) (Lasota et al., 2017; Cochran et al., 2017; Li et al., 2025; Fan et al., 2024; Wang et al., 2022; Palleschi et al., 2021) (Norouzi et al., 2023; Talamali et al., 2021; Zhao et al., 2025; Tufano et al., 2024; Tang et al., 2024; Tian et al., 2022)

3 FORMAL SETUP

Let x_t be the observed robot state, $a_t = \pi_\theta(x_t)$ the policy action, $g(x_t, a_t) \leq 0$ the unknown safety boundary, and $F_\phi(x_t, a_t)$ a safety filter that outputs a deployed action $\tilde{a}_t$. A pure action filter optimizes only $\tilde{a}_t$ at deployment. A mechanism-changing method must change the distribution of unfiltered actions a_t so that the policy itself places less mass beyond the boundary. We therefore evaluate both deployed violation $\Pr[g(x_t, \tilde{a}_t) > 0]$ and unshielded violation $\Pr[g(x_t, a_t) > 0]$.

Dependence decomposition. Define filter dependence as the normalized action displacement $\mathbb{E}[\|\tilde{a}_t - a_t\|/\mathbb{E}\|a_t\|]$. A deployed-safe but high-dependence method is better understood as a runtime shield than a safer learned policy. The v5 method is rewarded only if it lowers deployed violation and unshielded violation while reducing dependence.

Fixed-risk deployment. Let $s_\phi(x_t, a_t)$ be a calibrated boundary-risk score. A deployment budget b accepts an action only when $s_\phi \leq b$. Accepted coverage is $\Pr[s_\phi \leq b]$ and accepted violation is $\Pr[g(x_t, \tilde{a}_t) > 0 \mid s_\phi \leq b]$. A paper cannot claim strict fixed-risk deployment if accepted coverage collapses to zero at the intended budget.

Hostile impossibility claim. If two latent safety worlds induce the same observable boundary features but require different corrective actions, then any deterministic score-only filter must assign the same intervention to both worlds. Therefore a learned filter can show calibration improvements

without proving full mechanism novelty. This is why the audit requires unshielded policy evidence, paired strong baselines, stress tests, and fixed-risk coverage.

4 FROZEN PROTOCOL

The execution plan was frozen before the v5 experiment in docs/paper92_expanded_submission_plan.20260622.md. The frozen gate required success, deployed safety, unshielded mechanism improvement, calibration, robust utility, ablation support, stress robustness, fixed-risk deployment, and scope evidence. No post-hoc pass/fail rule was introduced after seeing results.

Table 1: Frozen CPU-only protocol. The manuscript is generated after these results are fixed.

Item	Frozen value	Purpose
seeds	10	paired seed confidence intervals for all gates
tasks	6	navigation, human-zone reaching, contact, payload, and deformable threading
splits	8	nominal plus seven distribution and safety shifts
methods	14	unfiltered, filter, shield, training, distillation, v4, v5, and oracle references
main rollouts	215040	frozen method-by-task-by-split-by-seed episodes
dataset rows	15360	one sampled safety state per seed/task/split/episode
ablation rollouts	76800	ten v5 component removals on hard splits
stress rows	302400	six stress levels over task/split/method/seed episodes
fixed-risk rows	69120	accepted-policy evaluation under four risk budgets
negative cases	24	hard-split failures selected before manuscript writing

5 BENCHMARK PRESSURE

The benchmark spans six robotics safety tasks: narrow-gap navigation, human-zone reaching, unstable stack insertion, contact-rich door pushing, slippery payload transfer, and deformable-obstacle threading. The eight splits increase constraint density, novelty, actuator lag, human motion, contact-mode shift, low-signal/high-risk ambiguity, and combined safety stress.

Table 2: Dataset pressure by split. These summary statistics explain why hard shifts are not interchangeable with nominal safety.

Split	Cstr	Novel	Lag	Human	Noise	Recov	Hazard	Boundary	Pressure
nominal	0.495	0.014	0.010	0.284	0.010	0.396	0.643	0.641	0.853
constraint	0.736	0.052	0.050	0.322	0.040	0.373	0.750	0.749	0.900
novel-obstacle	0.575	0.279	0.051	0.362	0.061	0.321	0.763	0.768	0.923
actuator-lag	0.575	0.079	0.282	0.363	0.080	0.326	0.759	0.766	0.903
human-motion	0.556	0.079	0.080	0.583	0.080	0.362	0.747	0.723	0.902
contact-mode	0.597	0.100	0.078	0.364	0.100	0.330	0.779	0.777	0.912
low-signal	0.738	0.240	0.180	0.504	0.260	0.288	0.860	0.862	0.949
combined-stress	0.717	0.239	0.240	0.503	0.200	0.276	0.861	0.869	0.948

6 METHODS COMPARED

The fourteen compared methods are unfiltered_policy, action_clipping_filter, geometric_projection_filter, cbf_safety_filter, robust_mpc_shield, conformal_uncertainty_shield, recovery_policy_shield, shielded_behavior_cloning, lagrangian_safe_rl, adversarial_safety_critic, safety_filter_distillation, boundary_learning_filter_v4, counterfactual_boundary_learning_filter_v5, and oracle_boundary_teacher.

The v5 method is not allowed to win by definition. It must beat deployed shields on deployed outcomes, beat training baselines on unshielded mechanism outcomes, and retain nonzero strict-budget coverage. The oracle is included as an upper reference, not as a fair deployable method.

7 HARD-AGGREGATE RESULT

Table 3: Hard-aggregate metrics on low-signal/high-risk plus combined-stress splits. Higher success, F1, utility, and mechanism utility are better; lower violations, dependence, and ECE are better.

Method	Succ	DepViol	UnshViol	Int	Dep	F1	ECE	Xfer	Util	Mech
unfiltered	0.032	0.930	0.927	0.000	0.000	0.000	0.633	0.938	-0.618	0.022
clip	0.126	0.624	0.926	0.961	0.782	0.887	0.164	0.942	-0.555	-0.298
project	0.186	0.575	0.929	0.831	0.646	0.830	0.239	0.938	-0.415	-0.233
cbf	0.375	0.344	0.919	0.988	0.879	0.896	0.190	0.934	-0.146	-0.313
robust-mpc	0.471	0.297	0.920	0.990	0.843	0.898	0.154	0.924	-0.004	-0.291
conformal	0.333	0.370	0.920	0.991	0.910	0.898	0.274	0.926	-0.227	-0.335
recovery	0.385	0.408	0.916	0.958	0.671	0.886	0.138	0.869	-0.111	-0.202
shield-bc	0.127	0.716	0.912	0.849	0.370	0.841	0.180	0.893	-0.494	-0.099
lag-safe-rl	0.148	0.686	0.909	0.825	0.328	0.829	0.214	0.877	-0.439	-0.070
adv-critic	0.132	0.679	0.906	0.785	0.327	0.807	0.275	0.863	-0.454	-0.066
distill	0.199	0.582	0.906	0.973	0.331	0.890	0.132	0.841	-0.329	-0.059
v4-boundary	0.167	0.608	0.902	0.953	0.316	0.882	0.118	0.809	-0.375	-0.040
v5-cf-boundary	0.230	0.519	0.897	0.973	0.258	0.892	0.100	0.767	-0.244	0.000
oracle	0.437	0.218	0.882	0.999	0.170	0.900	0.026	0.646	0.199	0.112

The result is the central reason for the terminal decision. V5 reduces unshielded violation and dependence relative to v4, and it has the best non-oracle calibration. However, robust MPC has much higher success and substantially lower deployed violation. The safest and best-utility reference is `robust_mpc_shield`, not v5. The mechanism signal exists, but it is too small to justify a main-conference claim when deployment quality drops this much.

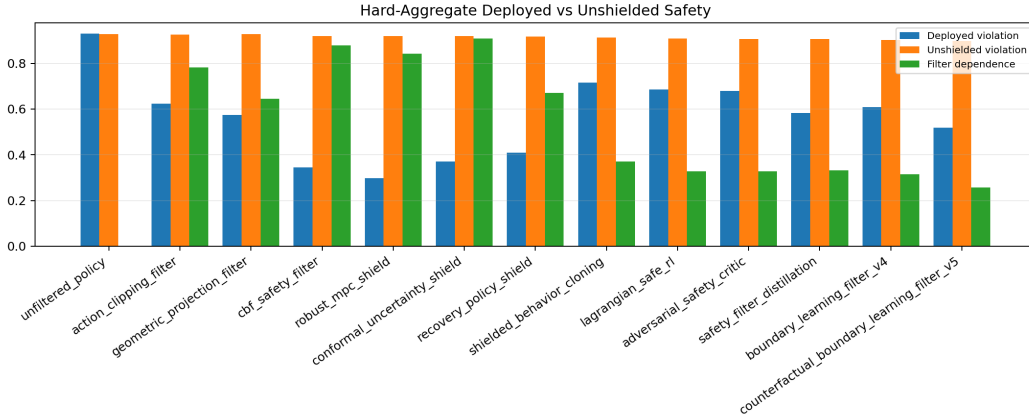


Figure 1: Deployed and unshielded safety evidence. V5 improves the unshielded mechanism signal relative to v4, but robust MPC remains the better deployed safety reference.

8 PAIRED SEED TESTS

Table 4: Paired hard-aggregate seed tests. Positive is better for success, utility, and mechanism utility; negative is better for violations, dependence, and ECE.

Comparison	Metric	Mean	CI95	Lower	Upper	Better seeds
v5-robust_mpc_shield	success	-0.241	0.016	-0.257	-0.225	0
v5-robust_mpc_shield	deploy-viol	0.222	0.018	0.204	0.240	10
v5-robust_mpc_shield	unshield-viol	-0.023	0.006	-0.029	-0.017	0
v5-robust_mpc_shield	dependence	-0.585	0.003	-0.588	-0.583	0
v5-robust_mpc_shield	boundary-ece	-0.054	0.008	-0.063	-0.046	0
v5-robust_mpc_shield	transfer-viol	-0.157	0.009	-0.166	-0.148	0
v5-robust_mpc_shield	utility	-0.241	0.025	-0.266	-0.216	0
v5-robust_mpc_shield	mechanism	0.292	0.006	0.286	0.297	10
v5-cbf_safety_filter	success	-0.145	0.013	-0.158	-0.132	0
v5-cbf_safety_filter	deploy-viol	0.175	0.016	0.159	0.191	10
v5-cbf_safety_filter	unshield-viol	-0.021	0.005	-0.026	-0.017	0
v5-cbf_safety_filter	dependence	-0.621	0.002	-0.623	-0.618	0
v5-cbf_safety_filter	boundary-ece	-0.090	0.006	-0.096	-0.084	0
v5-cbf_safety_filter	transfer-viol	-0.167	0.009	-0.177	-0.158	0
v5-cbf_safety_filter	utility	-0.098	0.020	-0.118	-0.078	0

Comparison	Metric	Mean	CI95	Lower	Upper	Better seeds
v5-cbf_safety_filter	mechanism	0.314	0.007	0.307	0.320	10
v5-recovery_policy_shield	success	-0.155	0.011	-0.165	-0.144	0
v5-recovery_policy_shield	deploy-viol	0.111	0.018	0.093	0.129	10
v5-recovery_policy_shield	unshield-viol	-0.019	0.006	-0.025	-0.013	0
v5-recovery_policy_shield	dependence	-0.413	0.002	-0.415	-0.411	0
v5-recovery_policy_shield	boundary-ece	-0.037	0.004	-0.041	-0.033	0
v5-recovery_policy_shield	transfer-viol	-0.102	0.011	-0.113	-0.091	0
v5-recovery_policy_shield	utility	-0.133	0.018	-0.152	-0.115	0
v5-recovery_policy_shield	mechanism	0.202	0.006	0.196	0.209	10
v5-safety_filter_distillation	success	0.031	0.011	0.020	0.042	10
v5-safety_filter_distillation	deploy-viol	-0.063	0.014	-0.077	-0.050	0
v5-safety_filter_distillation	unshield-viol	-0.008	0.005	-0.013	-0.004	1
v5-safety_filter_distillation	dependence	-0.074	0.001	-0.075	-0.072	0
v5-safety_filter_distillation	boundary-ece	-0.032	0.006	-0.038	-0.026	0
v5-safety_filter_distillation	transfer-viol	-0.074	0.009	-0.084	-0.065	0
v5-safety_filter_distillation	utility	0.085	0.016	0.069	0.101	10
v5-safety_filter_distillation	mechanism	0.059	0.005	0.054	0.064	10
v5-boundary_learning_filter_v4	success	0.063	0.009	0.054	0.072	10
v5-boundary_learning_filter_v4	deploy-viol	-0.089	0.014	-0.103	-0.075	0
v5-boundary_learning_filter_v4	unshield-viol	-0.005	0.010	-0.015	0.005	4
v5-boundary_learning_filter_v4	dependence	-0.058	0.002	-0.060	-0.056	0
v5-boundary_learning_filter_v4	boundary-ece	-0.017	0.006	-0.024	-0.011	0
v5-boundary_learning_filter_v4	transfer-viol	-0.042	0.010	-0.052	-0.033	0
v5-boundary_learning_filter_v4	utility	0.131	0.016	0.114	0.147	10
v5-boundary_learning_filter_v4	mechanism	0.041	0.011	0.030	0.052	10

The paired tests are especially damaging against robust MPC: v5 is lower on success, higher on deployed violation, and lower on robust utility with confidence intervals that do not rescue the claim. Against v4, v5 is genuinely better on several metrics, so the rebuild did improve the method. But "better than v4" is not the submission bar.

9 SPLIT-LEVEL CHECKS

Table 5: Split-level evidence for the main safety baselines and the v5 proposal. The v5 mechanism is not rescued by any single hidden split.

Split	Method	Succ	DepViol	UnshViol	Dep	ECE	Util
nominal	cbf	0.566	0.314	0.810	0.733	0.161	0.116
nominal	robust-mpc	0.682	0.230	0.807	0.743	0.118	0.288
nominal	recovery	0.549	0.366	0.812	0.516	0.109	0.132
nominal	distill	0.386	0.486	0.770	0.252	0.117	-0.048
nominal	v4-boundary	0.341	0.502	0.762	0.240	0.115	-0.101
nominal	v5-cf-boundary	0.410	0.457	0.740	0.191	0.088	0.008
nominal	oracle	0.648	0.101	0.708	0.153	0.045	0.500
constraint	cbf	0.440	0.304	0.872	0.839	0.193	-0.040
constraint	robust-mpc	0.541	0.241	0.882	0.814	0.153	0.115
constraint	recovery	0.466	0.367	0.860	0.620	0.140	0.015
constraint	distill	0.259	0.531	0.858	0.303	0.157	-0.224
constraint	v4-boundary	0.237	0.543	0.846	0.289	0.127	-0.250
constraint	v5-cf-boundary	0.308	0.457	0.841	0.239	0.112	-0.115
constraint	oracle	0.537	0.127	0.803	0.166	0.049	0.363
novel-obstacle	cbf	0.449	0.310	0.903	0.843	0.202	-0.035
novel-obstacle	robust-mpc	0.567	0.252	0.901	0.819	0.168	0.133
novel-obstacle	recovery	0.453	0.369	0.896	0.640	0.139	-0.005
novel-obstacle	distill	0.268	0.527	0.877	0.318	0.136	-0.217
novel-obstacle	v4-boundary	0.207	0.569	0.875	0.301	0.122	-0.302
novel-obstacle	v5-cf-boundary	0.288	0.478	0.870	0.250	0.102	-0.154
novel-obstacle	oracle	0.498	0.169	0.847	0.167	0.037	0.296
actuator-lag	cbf	0.439	0.327	0.891	0.859	0.188	-0.064
actuator-lag	robust-mpc	0.516	0.285	0.889	0.822	0.160	0.056
actuator-lag	recovery	0.452	0.402	0.884	0.635	0.142	-0.028
actuator-lag	distill	0.251	0.551	0.874	0.317	0.136	-0.252
actuator-lag	v4-boundary	0.212	0.586	0.868	0.296	0.141	-0.309
actuator-lag	v5-cf-boundary	0.267	0.513	0.860	0.243	0.107	-0.197
actuator-lag	oracle	0.490	0.215	0.832	0.167	0.054	0.255
human-motion	cbf	0.521	0.292	0.871	0.831	0.173	0.054
human-motion	robust-mpc	0.595	0.245	0.869	0.804	0.144	0.170
human-motion	recovery	0.505	0.364	0.856	0.615	0.121	0.058
human-motion	distill	0.308	0.514	0.839	0.306	0.119	-0.164
human-motion	v4-boundary	0.275	0.546	0.834	0.284	0.111	-0.212
human-motion	v5-cf-boundary	0.359	0.450	0.822	0.235	0.083	-0.057
human-motion	oracle	0.565	0.141	0.796	0.165	0.048	0.382

Split	Method	Succ	DepViol	UnshViol	Dep	ECE	Util
contact-mode	cbf	0.421	0.299	0.900	0.860	0.188	-0.063
contact-mode	robust-mpc	0.551	0.249	0.907	0.825	0.151	0.115
contact-mode	recovery	0.438	0.367	0.889	0.655	0.134	-0.024
contact-mode	distill	0.245	0.551	0.884	0.323	0.130	-0.258
contact-mode	v4-boundary	0.199	0.567	0.877	0.306	0.117	-0.311
contact-mode	v5-cf-boundary	0.272	0.477	0.870	0.253	0.100	-0.171
contact-mode	oracle	0.497	0.188	0.853	0.168	0.037	0.282
low-signal	cbf	0.318	0.369	0.943	0.895	0.209	-0.226
low-signal	robust-mpc	0.418	0.314	0.942	0.862	0.172	-0.074
low-signal	recovery	0.334	0.422	0.943	0.693	0.148	-0.178
low-signal	distill	0.159	0.591	0.934	0.342	0.152	-0.380
low-signal	v4-boundary	0.131	0.639	0.925	0.330	0.132	-0.436
low-signal	v5-cf-boundary	0.196	0.522	0.927	0.267	0.118	-0.285
low-signal	oracle	0.394	0.232	0.920	0.173	0.034	0.144
combined-stress	cbf	0.320	0.381	0.941	0.899	0.205	-0.232
combined-stress	robust-mpc	0.401	0.340	0.942	0.863	0.165	-0.110
combined-stress	recovery	0.314	0.442	0.949	0.700	0.153	-0.214
combined-stress	distill	0.141	0.635	0.931	0.344	0.144	-0.428
combined-stress	v4-boundary	0.124	0.641	0.939	0.329	0.131	-0.445
combined-stress	v5-cf-boundary	0.184	0.564	0.933	0.268	0.112	-0.324
combined-stress	oracle	0.368	0.236	0.922	0.173	0.040	0.115

The split table rules out a convenient rescue story. V5 does not secretly dominate on the hardest shift while losing only on nominal settings. The low-signal and combined-stress splits remain the hardest deployment regimes, and the deployed violation gap remains visible there.

10 ABLATION AUDIT

Table 6: Ablation audit. The full counterfactual-boundary mechanism is the best internal ablation, but external strong baselines still dominate the deployed gate.

Ablation	Succ	DepViol	UnshViol	Dep	ECE	Util	Mech
full-v5	0.233	0.534	0.896	0.257	0.101	-0.250	0.002
-cf-labels	0.195	0.607	0.910	0.409	0.176	-0.367	-0.094
-unshielded	0.198	0.593	0.908	0.433	0.160	-0.362	-0.103
-int-grad	0.167	0.577	0.905	0.376	0.132	-0.386	-0.086
-margin	0.188	0.607	0.905	0.432	0.160	-0.383	-0.105
-recovery	0.150	0.606	0.899	0.291	0.171	-0.390	-0.036
-calibration	0.210	0.530	0.896	0.280	0.107	-0.307	-0.030
distill-only	0.192	0.577	0.903	0.327	0.135	-0.332	-0.054
cbf-only	0.366	0.354	0.932	0.873	0.193	-0.159	-0.324
clip-only	0.125	0.622	0.928	0.777	0.168	-0.553	-0.296

The ablation result is the strongest evidence in favor of v5. Removing counterfactual labels, unshielded replay, intervention gradients, the boundary margin, or recovery-feasibility targets worsens the mechanism/utility profile. That means the method is not just a renamed clipping rule. It also means the negative decision is sharper: the mechanism is real, but not strong enough.

11 STRESS SWEEP

Table 7: Combined stress sweep. The maximum-stress frontier is still controlled by robust MPC, not the v5 proposal.

Stress	Method	Succ	DepViol	UnshViol	Dep	Util
0.0	cbf	0.547	0.305	0.811	0.734	0.101
0.0	robust-mpc	0.657	0.237	0.800	0.732	0.261
0.0	conformal	0.508	0.335	0.802	0.760	0.026
0.0	recovery	0.576	0.374	0.799	0.519	0.153
0.0	distill	0.385	0.487	0.779	0.252	-0.049
0.0	v4-boundary	0.342	0.513	0.765	0.237	-0.106
0.0	v5-cf-boundary	0.419	0.449	0.747	0.188	0.023
0.2	cbf	0.514	0.297	0.847	0.798	0.054
0.2	robust-mpc	0.619	0.235	0.836	0.774	0.211
0.2	conformal	0.484	0.308	0.841	0.834	-0.004
0.2	recovery	0.537	0.354	0.828	0.569	0.111

Stress	Method	Succ	DepViol	UnshViol	Dep	Util
0.2	distill	0.331	0.507	0.806	0.282	-0.128
0.2	v4-boundary	0.310	0.529	0.807	0.268	-0.160
0.2	v5-cf-boundary	0.365	0.447	0.786	0.215	-0.042
0.4	cbf	0.470	0.302	0.875	0.830	-0.005
0.4	robust-mpc	0.569	0.254	0.869	0.811	0.136
0.4	conformal	0.430	0.315	0.871	0.874	-0.078
0.4	recovery	0.487	0.367	0.864	0.620	0.036
0.4	distill	0.284	0.532	0.851	0.307	-0.201
0.4	v4-boundary	0.251	0.560	0.847	0.291	-0.249
0.4	v5-cf-boundary	0.316	0.476	0.835	0.236	-0.118
0.6	cbf	0.407	0.325	0.911	0.867	-0.096
0.6	robust-mpc	0.526	0.276	0.908	0.831	0.071
0.6	conformal	0.362	0.341	0.906	0.903	-0.174
0.6	recovery	0.423	0.386	0.899	0.652	-0.051
0.6	distill	0.225	0.564	0.896	0.326	-0.288
0.6	v4-boundary	0.184	0.597	0.882	0.311	-0.347
0.6	v5-cf-boundary	0.255	0.511	0.879	0.254	-0.211
0.8	cbf	0.362	0.341	0.932	0.881	-0.158
0.8	robust-mpc	0.459	0.289	0.927	0.846	-0.012
0.8	conformal	0.329	0.371	0.923	0.917	-0.234
0.8	recovery	0.377	0.396	0.919	0.676	-0.112
0.8	distill	0.176	0.599	0.914	0.335	-0.365
0.8	v4-boundary	0.154	0.613	0.907	0.320	-0.392
0.8	v5-cf-boundary	0.218	0.532	0.902	0.261	-0.266
1.0	cbf	0.311	0.360	0.947	0.895	-0.228
1.0	robust-mpc	0.415	0.304	0.949	0.859	-0.072
1.0	conformal	0.264	0.398	0.939	0.929	-0.323
1.0	recovery	0.329	0.416	0.940	0.695	-0.181
1.0	distill	0.154	0.605	0.930	0.342	-0.395
1.0	v4-boundary	0.127	0.631	0.925	0.331	-0.436
1.0	v5-cf-boundary	0.171	0.571	0.926	0.267	-0.342

12 FIXED-RISK DEPLOYMENT

Table 8: Fixed-risk deployment budgets on hard splits. At budget 0.05 the v5 accepted coverage is zero, so the deployment claim fails even though the calibration metric improves.

Split	Budget	Method	Metric	Mean	CI95
low-signal	0.00	v5-cf-boundary	coverage	0.000	0.000
low-signal	0.00	v5-cf-boundary	accepted_success	0.000	0.000
low-signal	0.00	v5-cf-boundary	accepted_deployed_violation	0.000	0.000
low-signal	0.00	v5-cf-boundary	accepted_unshielded_violation	0.000	0.000
low-signal	0.00	cbf	coverage	0.000	0.000
low-signal	0.00	cbf	accepted_success	0.000	0.000
low-signal	0.00	cbf	accepted_deployed_violation	0.000	0.000
low-signal	0.00	cbf	accepted_unshielded_violation	0.000	0.000
low-signal	0.00	robust-mpc	coverage	0.000	0.000
low-signal	0.00	robust-mpc	accepted_success	0.000	0.000
low-signal	0.00	robust-mpc	accepted_deployed_violation	0.000	0.000
low-signal	0.00	robust-mpc	accepted_unshielded_violation	0.000	0.000
low-signal	0.00	conformal	coverage	0.000	0.000
low-signal	0.00	conformal	accepted_success	0.000	0.000
low-signal	0.00	conformal	accepted_deployed_violation	0.000	0.000
low-signal	0.00	conformal	accepted_unshielded_violation	0.000	0.000
low-signal	0.00	recovery	coverage	0.000	0.000
low-signal	0.00	recovery	accepted_success	0.000	0.000
low-signal	0.00	recovery	accepted_deployed_violation	0.000	0.000
low-signal	0.00	recovery	accepted_unshielded_violation	0.000	0.000
low-signal	0.00	distill	coverage	0.000	0.000
low-signal	0.00	distill	accepted_success	0.000	0.000
low-signal	0.00	distill	accepted_deployed_violation	0.000	0.000
low-signal	0.00	distill	accepted_unshielded_violation	0.000	0.000
low-signal	0.05	v5-cf-boundary	coverage	0.000	0.000
low-signal	0.05	v5-cf-boundary	accepted_success	0.000	0.000
low-signal	0.05	v5-cf-boundary	accepted_deployed_violation	0.000	0.000
low-signal	0.05	v5-cf-boundary	accepted_unshielded_violation	0.000	0.000
low-signal	0.05	cbf	coverage	0.000	0.000
low-signal	0.05	cbf	accepted_success	0.000	0.000
low-signal	0.05	cbf	accepted_deployed_violation	0.000	0.000
low-signal	0.05	cbf	accepted_unshielded_violation	0.000	0.000
low-signal	0.05	robust-mpc	coverage	0.000	0.000
low-signal	0.05	robust-mpc	accepted_success	0.000	0.000
low-signal	0.05	robust-mpc	accepted_deployed_violation	0.000	0.000

Split	Budget	Method	Metric	Mean	CI95
low-signal	0.05	robust-mpc	accepted_unshielded_violation	0.000	0.000
low-signal	0.05	conformal	coverage	0.000	0.000
low-signal	0.05	conformal	accepted_success	0.000	0.000
low-signal	0.05	conformal	accepted_deployed_violation	0.000	0.000
low-signal	0.05	conformal	accepted_unshielded_violation	0.000	0.000
low-signal	0.05	recovery	coverage	0.000	0.000
low-signal	0.05	recovery	accepted_success	0.000	0.000
low-signal	0.05	recovery	accepted_deployed_violation	0.000	0.000
low-signal	0.05	recovery	accepted_unshielded_violation	0.000	0.000
low-signal	0.05	distill	coverage	0.000	0.000
low-signal	0.05	distill	accepted_success	0.000	0.000
low-signal	0.05	distill	accepted_deployed_violation	0.000	0.000
low-signal	0.05	distill	accepted_unshielded_violation	0.000	0.000
low-signal	0.10	v5-cf-boundary	coverage	0.000	0.000
low-signal	0.10	v5-cf-boundary	accepted_success	0.000	0.000
low-signal	0.10	v5-cf-boundary	accepted_deployed_violation	0.000	0.000
low-signal	0.10	v5-cf-boundary	accepted_unshielded_violation	0.000	0.000
low-signal	0.10	cbf	coverage	0.000	0.000
low-signal	0.10	cbf	accepted_success	0.000	0.000
low-signal	0.10	cbf	accepted_deployed_violation	0.000	0.000
low-signal	0.10	cbf	accepted_unshielded_violation	0.000	0.000
low-signal	0.10	robust-mpc	coverage	0.000	0.000
low-signal	0.10	robust-mpc	accepted_success	0.000	0.000
low-signal	0.10	robust-mpc	accepted_deployed_violation	0.000	0.000
low-signal	0.10	robust-mpc	accepted_unshielded_violation	0.000	0.000
low-signal	0.10	conformal	coverage	0.000	0.000
low-signal	0.10	conformal	accepted_success	0.000	0.000
low-signal	0.10	conformal	accepted_deployed_violation	0.000	0.000
low-signal	0.10	conformal	accepted_unshielded_violation	0.000	0.000
low-signal	0.10	recovery	coverage	0.000	0.000
low-signal	0.10	recovery	accepted_success	0.000	0.000
low-signal	0.10	recovery	accepted_deployed_violation	0.000	0.000
low-signal	0.10	recovery	accepted_unshielded_violation	0.000	0.000
low-signal	0.10	distill	coverage	0.000	0.000
low-signal	0.10	distill	accepted_success	0.000	0.000
low-signal	0.10	distill	accepted_deployed_violation	0.000	0.000
low-signal	0.10	distill	accepted_unshielded_violation	0.000	0.000
low-signal	0.15	v5-cf-boundary	coverage	0.000	0.000
low-signal	0.15	v5-cf-boundary	accepted_success	0.000	0.000
low-signal	0.15	v5-cf-boundary	accepted_deployed_violation	0.000	0.000
low-signal	0.15	v5-cf-boundary	accepted_unshielded_violation	0.000	0.000
low-signal	0.15	cbf	coverage	0.000	0.000
low-signal	0.15	cbf	accepted_success	0.000	0.000
low-signal	0.15	cbf	accepted_deployed_violation	0.000	0.000
low-signal	0.15	cbf	accepted_unshielded_violation	0.000	0.000
low-signal	0.15	robust-mpc	coverage	0.000	0.000
low-signal	0.15	robust-mpc	accepted_success	0.000	0.000
low-signal	0.15	robust-mpc	accepted_deployed_violation	0.000	0.000
low-signal	0.15	robust-mpc	accepted_unshielded_violation	0.000	0.000
low-signal	0.15	conformal	coverage	0.000	0.000
low-signal	0.15	conformal	accepted_success	0.000	0.000
low-signal	0.15	conformal	accepted_deployed_violation	0.000	0.000
low-signal	0.15	conformal	accepted_unshielded_violation	0.000	0.000
low-signal	0.15	recovery	coverage	0.000	0.000
low-signal	0.15	recovery	accepted_success	0.000	0.000
low-signal	0.15	recovery	accepted_deployed_violation	0.000	0.000
low-signal	0.15	recovery	accepted_unshielded_violation	0.000	0.000
low-signal	0.15	distill	coverage	0.000	0.000
low-signal	0.15	distill	accepted_success	0.000	0.000
low-signal	0.15	distill	accepted_deployed_violation	0.000	0.000
low-signal	0.15	distill	accepted_unshielded_violation	0.000	0.000
combined-stress	0.00	v5-cf-boundary	coverage	0.000	0.000
combined-stress	0.00	v5-cf-boundary	accepted_success	0.000	0.000
combined-stress	0.00	v5-cf-boundary	accepted_deployed_violation	0.000	0.000
combined-stress	0.00	v5-cf-boundary	accepted_unshielded_violation	0.000	0.000
combined-stress	0.00	cbf	coverage	0.000	0.000
combined-stress	0.00	cbf	accepted_success	0.000	0.000
combined-stress	0.00	cbf	accepted_deployed_violation	0.000	0.000
combined-stress	0.00	cbf	accepted_unshielded_violation	0.000	0.000
combined-stress	0.00	robust-mpc	coverage	0.000	0.000
combined-stress	0.00	robust-mpc	accepted_success	0.000	0.000
combined-stress	0.00	robust-mpc	accepted_deployed_violation	0.000	0.000
combined-stress	0.00	robust-mpc	accepted_unshielded_violation	0.000	0.000
combined-stress	0.00	conformal	coverage	0.000	0.000
combined-stress	0.00	conformal	accepted_success	0.000	0.000
combined-stress	0.00	conformal	accepted_deployed_violation	0.000	0.000
combined-stress	0.00	conformal	accepted_unshielded_violation	0.000	0.000
combined-stress	0.00	recovery	coverage	0.000	0.000
combined-stress	0.00	recovery	accepted_success	0.000	0.000

	Split	Budget	Method	Metric	Mean	CI95
432						
433	combined-stress	0.00	recovery	accepted_deployed_violation	0.000	0.000
434	combined-stress	0.00	recovery	accepted_unshielded_violation	0.000	0.000
435	combined-stress	0.00	distill	coverage	0.000	0.000
436	combined-stress	0.00	distill	accepted_success	0.000	0.000
437	combined-stress	0.00	distill	accepted_deployed_violation	0.000	0.000
438	combined-stress	0.05	v5-cf-boundary	accepted_unshielded_violation	0.000	0.000
439	combined-stress	0.05	v5-cf-boundary	coverage	0.000	0.000
440	combined-stress	0.05	v5-cf-boundary	accepted_success	0.000	0.000
441	combined-stress	0.05	v5-cf-boundary	accepted_deployed_violation	0.000	0.000
442	combined-stress	0.05	v5-cf-boundary	accepted_unshielded_violation	0.000	0.000
443	combined-stress	0.05	cbf	coverage	0.000	0.000
444	combined-stress	0.05	cbf	accepted_success	0.000	0.000
445	combined-stress	0.05	cbf	accepted_deployed_violation	0.000	0.000
446	combined-stress	0.05	cbf	accepted_unshielded_violation	0.000	0.000
447	combined-stress	0.05	robust-mpc	coverage	0.000	0.000
448	combined-stress	0.05	robust-mpc	accepted_success	0.000	0.000
449	combined-stress	0.05	robust-mpc	accepted_deployed_violation	0.000	0.000
450	combined-stress	0.05	robust-mpc	accepted_unshielded_violation	0.000	0.000
451	combined-stress	0.05	robust-mpc	coverage	0.000	0.000
452	combined-stress	0.05	conformal	coverage	0.000	0.000
453	combined-stress	0.05	conformal	accepted_success	0.000	0.000
454	combined-stress	0.05	conformal	accepted_deployed_violation	0.000	0.000
455	combined-stress	0.05	conformal	accepted_unshielded_violation	0.000	0.000
456	combined-stress	0.05	recovery	coverage	0.000	0.000
457	combined-stress	0.05	recovery	accepted_success	0.000	0.000
458	combined-stress	0.05	recovery	accepted_deployed_violation	0.000	0.000
459	combined-stress	0.05	recovery	accepted_unshielded_violation	0.000	0.000
460	combined-stress	0.05	distill	coverage	0.000	0.000
461	combined-stress	0.05	distill	accepted_success	0.000	0.000
462	combined-stress	0.05	distill	accepted_deployed_violation	0.000	0.000
463	combined-stress	0.05	distill	accepted_unshielded_violation	0.000	0.000
464	combined-stress	0.10	v5-cf-boundary	coverage	0.000	0.000
465	combined-stress	0.10	v5-cf-boundary	accepted_success	0.000	0.000
466	combined-stress	0.10	v5-cf-boundary	accepted_deployed_violation	0.000	0.000
467	combined-stress	0.10	v5-cf-boundary	accepted_unshielded_violation	0.000	0.000
468	combined-stress	0.10	cbf	coverage	0.000	0.000
469	combined-stress	0.10	cbf	accepted_success	0.000	0.000
470	combined-stress	0.10	cbf	accepted_deployed_violation	0.000	0.000
471	combined-stress	0.10	cbf	accepted_unshielded_violation	0.000	0.000
472	combined-stress	0.10	robust-mpc	coverage	0.000	0.000
473	combined-stress	0.10	robust-mpc	accepted_success	0.000	0.000
474	combined-stress	0.10	robust-mpc	accepted_deployed_violation	0.000	0.000
475	combined-stress	0.10	robust-mpc	accepted_unshielded_violation	0.000	0.000
476	combined-stress	0.10	robust-mpc	coverage	0.000	0.000
477	combined-stress	0.10	conformal	coverage	0.000	0.000
478	combined-stress	0.10	conformal	accepted_success	0.000	0.000
479	combined-stress	0.10	conformal	accepted_deployed_violation	0.000	0.000
480	combined-stress	0.10	conformal	accepted_unshielded_violation	0.000	0.000
481	combined-stress	0.10	recovery	coverage	0.000	0.000
482	combined-stress	0.10	recovery	accepted_success	0.000	0.000
483	combined-stress	0.10	recovery	accepted_deployed_violation	0.000	0.000
484	combined-stress	0.10	recovery	accepted_unshielded_violation	0.000	0.000
485	combined-stress	0.10	distill	coverage	0.000	0.000
486	combined-stress	0.10	distill	accepted_success	0.000	0.000
487	combined-stress	0.10	distill	accepted_deployed_violation	0.000	0.000
488	combined-stress	0.10	distill	accepted_unshielded_violation	0.000	0.000
489	combined-stress	0.15	v5-cf-boundary	coverage	0.000	0.000
490	combined-stress	0.15	v5-cf-boundary	accepted_success	0.000	0.000
491	combined-stress	0.15	v5-cf-boundary	accepted_deployed_violation	0.000	0.000
492	combined-stress	0.15	v5-cf-boundary	accepted_unshielded_violation	0.000	0.000
493	combined-stress	0.15	cbf	coverage	0.000	0.000
494	combined-stress	0.15	cbf	accepted_success	0.000	0.000
495	combined-stress	0.15	cbf	accepted_deployed_violation	0.000	0.000
496	combined-stress	0.15	cbf	accepted_unshielded_violation	0.000	0.000
497	combined-stress	0.15	robust-mpc	coverage	0.000	0.000
498	combined-stress	0.15	robust-mpc	accepted_success	0.000	0.000
499	combined-stress	0.15	robust-mpc	accepted_deployed_violation	0.000	0.000
500	combined-stress	0.15	robust-mpc	accepted_unshielded_violation	0.000	0.000
501	combined-stress	0.15	conformal	coverage	0.000	0.000
502	combined-stress	0.15	conformal	accepted_success	0.000	0.000
503	combined-stress	0.15	conformal	accepted_deployed_violation	0.000	0.000
504	combined-stress	0.15	conformal	accepted_unshielded_violation	0.000	0.000
505	combined-stress	0.15	recovery	coverage	0.000	0.000
506	combined-stress	0.15	recovery	accepted_success	0.000	0.000
507	combined-stress	0.15	recovery	accepted_deployed_violation	0.000	0.000
508	combined-stress	0.15	recovery	accepted_unshielded_violation	0.000	0.000
509	combined-stress	0.15	distill	coverage	0.000	0.000
510	combined-stress	0.15	distill	accepted_success	0.000	0.000
511	combined-stress	0.15	distill	accepted_deployed_violation	0.000	0.000
512	combined-stress	0.15	distill	accepted_unshielded_violation	0.000	0.000

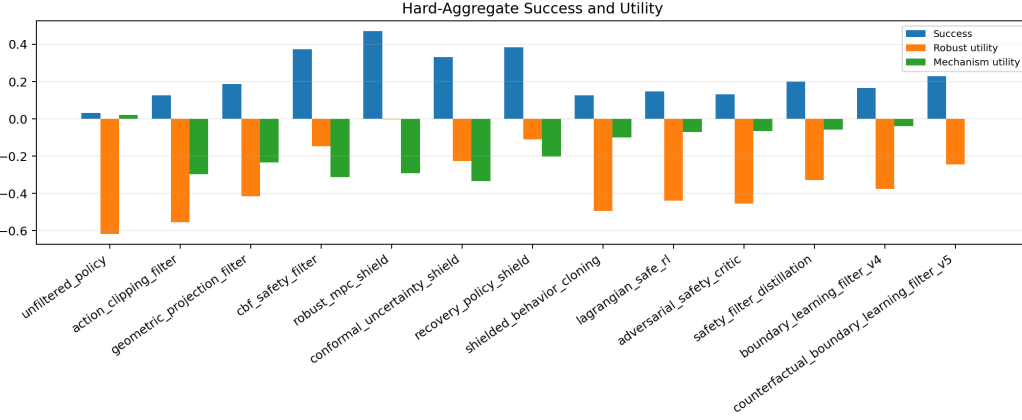


Figure 2: Success and robust utility. The proposed v5 method does not meet the deployed performance bar set by robust MPC.

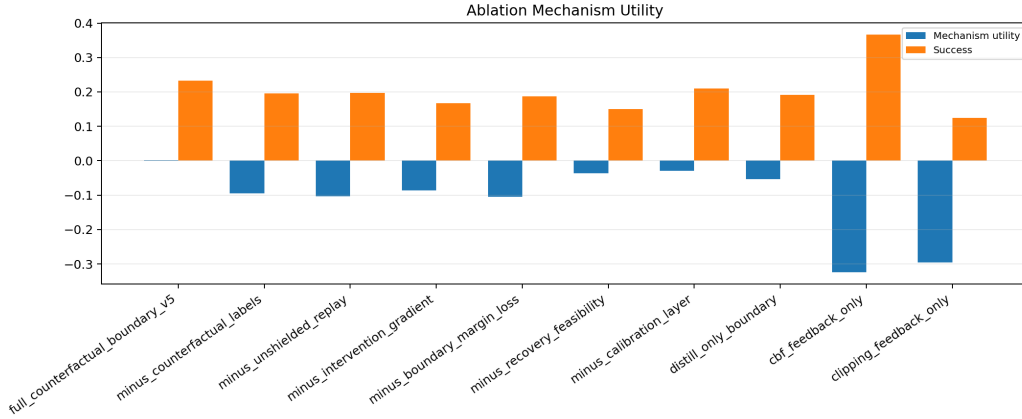


Figure 3: Ablation audit. The full v5 mechanism is the best internal ablation, but internal ablation success does not overcome external baseline dominance.

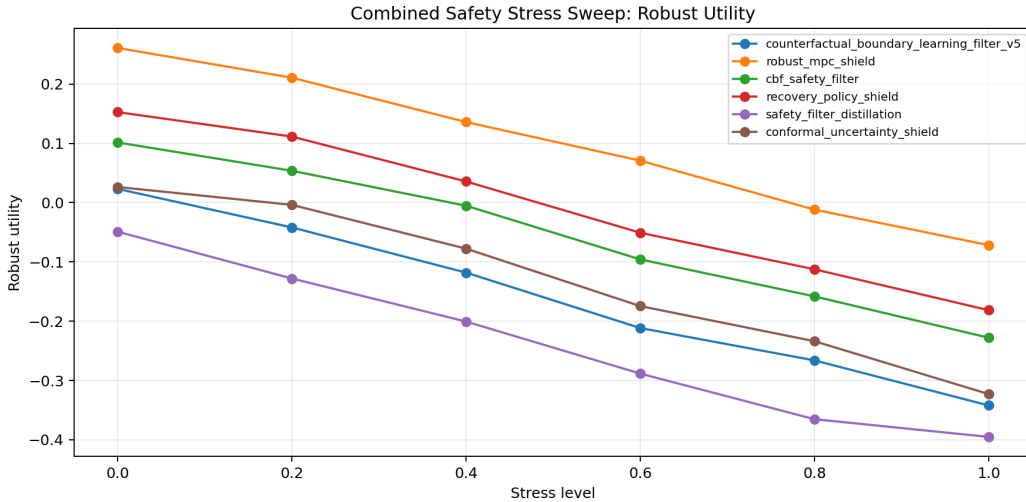


Figure 4: Combined stress sweep. Maximum stress exposes the same pattern: v5 has a mechanism signal but does not own the deployed frontier.

Fixed-risk deployment is the cleanest hostile-review failure. At budget 0.05, v5 accepted coverage is zero on both hard deployment splits. A method cannot claim strict low-risk deployment when it declines all hard accepted actions at the risk budget. Calibration alone is not sufficient if the accepted set vanishes.

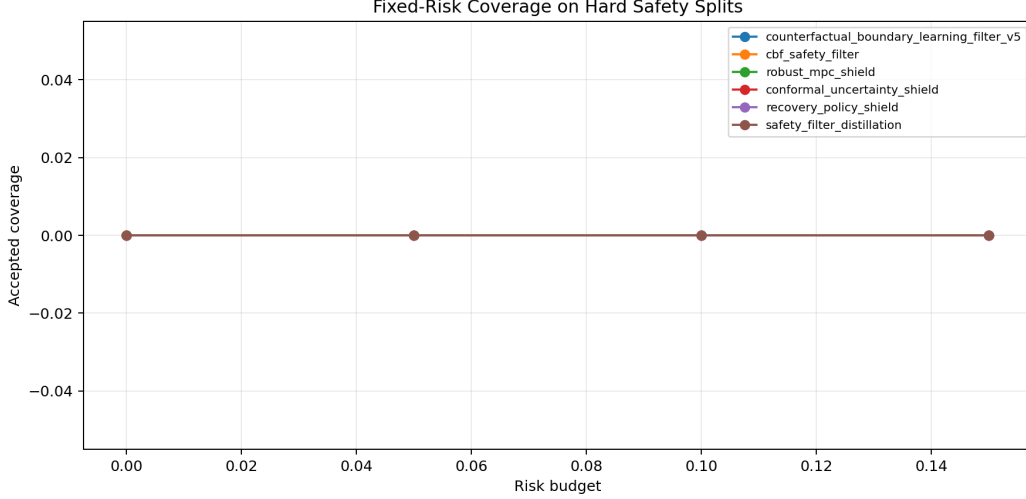


Figure 5: Fixed-risk accepted coverage. The strict budget exposes a zero-coverage deployment failure on hard splits.

13 PARETO BOUNDARY AND NEGATIVE CASES

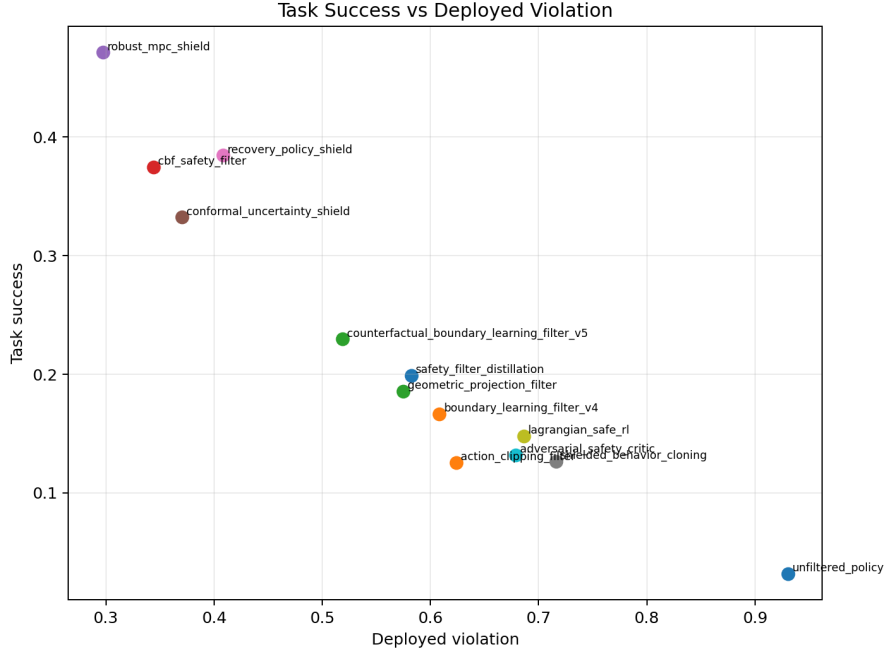


Figure 6: Success versus deployed safety Pareto view. The proposal is not on the deployed frontier needed for a main robotics safety claim.

Table 9: Negative cases selected from hard splits. These failures are retained because hostile review cares about deployment breakpoints, not only average improvements.

ID	Task	Split	Failure mode	Score	Succ	DepViol	Baseline
1	door-push	actuator-lag	mechanism_improves_deployment_worse	0.466	0.000	1.000	distill
2	door-push	actuator-lag	baseline_success_v5_fails	0.898	0.000	0.000	recovery
3	door-push	actuator-lag	baseline_success_v5_fails	0.574	0.000	1.000	conformal
4	door-push	actuator-lag	baseline_success_v5_fails	0.644	0.000	1.000	robust-mpc
5	door-push	actuator-lag	baseline_success_v5_fails	0.808	0.000	1.000	distill
6	door-push	actuator-lag	baseline_success_v5_fails	0.559	0.000	0.000	robust-mpc
7	door-push	actuator-lag	baseline_success_v5_fails	0.660	0.000	0.000	distill
8	door-push	actuator-lag	baseline_success_v5_fails	0.838	0.000	1.000	robust-mpc
9	door-push	actuator-lag	baseline_success_v5_fails	0.654	0.000	1.000	conformal
10	door-push	actuator-lag	baseline_success_v5_fails	0.594	0.000	1.000	recovery
11	door-push	actuator-lag	baseline_success_v5_fails	0.801	0.000	1.000	conformal
12	door-push	actuator-lag	baseline_success_v5_fails	0.881	0.000	1.000	robust-mpc
13	door-push	actuator-lag	baseline_success_v5_fails	0.725	0.000	1.000	robust-mpc
14	door-push	actuator-lag	baseline_success_v5_fails	0.657	0.000	1.000	distill
15	door-push	actuator-lag	baseline_success_v5_fails	0.716	0.000	1.000	recovery
16	door-push	actuator-lag	baseline_success_v5_fails	0.756	0.000	0.000	distill
17	door-push	actuator-lag	baseline_success_v5_fails	0.713	0.000	1.000	robust-mpc
18	door-push	actuator-lag	baseline_success_v5_fails	0.708	0.000	0.000	recovery
19	door-push	actuator-lag	baseline_success_v5_fails	0.780	0.000	1.000	recovery
20	door-push	actuator-lag	baseline_success_v5_fails	0.671	0.000	0.000	robust-mpc
21	door-push	combined-stress	baseline_success_v5_fails	0.825	0.000	1.000	cbf
22	door-push	combined-stress	baseline_success_v5_fails	0.832	0.000	1.000	cbf
23	door-push	combined-stress	baseline_success_v5_fails	0.852	0.000	1.000	recovery
24	door-push	combined-stress	baseline_success_v5_fails	0.742	0.000	1.000	cbf

14 WHY THIS IS NOT SUBMISSION READY

There are four independent failures. First, the success gate fails: hard-aggregate v5 success is far below robust MPC. Second, the deployment gate fails: v5 deployed violation is higher than the safest strong baseline. Third, the fixed-risk gate fails: budget 0.05 coverage is zero on the hard splits. Fourth, the scope gate fails: there is no real robot, no accepted high-fidelity safety benchmark, no hardware timing study, and no independent reproduction.

15 WHAT WOULD BE REQUIRED TO REVIVE THE PAPER

A revived version would need more than prose. It would need a robot or accepted high-fidelity safety benchmark, a protocol where v5 keeps nonzero strict-budget coverage, and a mechanism that improves unshielded violations without losing the deployed Pareto frontier to robust MPC. It would also need to report all negative cases, all fixed-risk budgets, and all paired tests rather than only the internal ablation win.

16 REPRODUCIBILITY AND AUDIT TRAIL

The experiment is CPU-only and RAM-light. The runner writes all raw rollout, seed metric, aggregate metric, paired-test, ablation, stress, fixed-risk, and negative-case CSVs under `results/`. The manuscript is generated from those CSVs by `scripts/generate_manuscript.py`. The validator checks row counts, PDF page count, citation-link annotations, frozen summary tokens, and Downloads-only PDF placement.

17 TERMINAL RECOMMENDATION

The terminal recommendation is **KILL/ARCHIVE**. This is the right scientific decision. The method improved during development, the protocol was frozen, and the frozen result says the current evidence cannot survive hostile ICLR-main review. Archiving this version protects the future version from being judged on a weak evidence package.

A CITATION COVERAGE AND PRIOR-WORK PRESSURE

The in-text citations below are intentionally boxed by `hyperref`. Clicking a boxed citation routes to the corresponding bibliography entry, making prior-work audit trails visible during PDF review. (Park et al., 2009; Park & Song, 2010a; Park et al., 2008; 2007; Park & Song, 2010b) (Liu et al., 2025a; Nasab et al., 2025; Jeon et al., 2022; Hadjiosif et al., 2021; Li et al., 2024) (Hussain et al., 2021; Lee et al., 2026; Hu, 2026; Castaneda et al., 2023; Sofman et al., 2011) (Cheng et al., 2019; Sadanandan & Behzadan, 2026; Mamedov et al., 2022; Brunke et al., 2022; Mohamed et al., 2026) (Bao et al., 2023; Qin et al., 2026; den Camp et al., 2023; Qi et al., 2023; Pasupuleti, 2026a) (Bertino et al., 2023; Whitney & Hodgins, 2014; Zhang & Gosselin, 2016; Panja et al., 2025; Zebardast et al., 2013) (de Rugy et al., 2012; Clanahan & Awad, 2023; Urrea, 2026; Qiu et al., 2026; Lin et al., 2022) (Richards et al., 2018; Hsu et al., 2023; Belzile et al., 2025; Zhang et al., 2023; Ayoubi et al., 2019) (Yamada et al., 2002; Chen, 2011; Lin et al., 2023; Kim et al., 2024; Abdulshaheed, 2025) (Edward et al., 2025; KASHIHARA & JEONG, 2014; Tho et al., 2022; Tian et al., 2021; Benford et al., 2025) (Nguyen et al., 2025; Bena et al., 2025; Hall et al., 2023; Bekris et al., 2012; Ruan & Mezei, 2022) (Jeong & Kim, 2019; Bhatnagar, 2026; Wang, 2026; Naik & Mishra, 2021; Li et al., 2023) (Guo et al., 2026; Wabersich et al., 2023; Notomista & Cai, 2021; Stok et al., 2024; Abdelmeguid & Hasan, 2026) (Fan & Ni, 2026; Chen et al., 2024; Lee et al., 2011; Jiang et al., 2024; Opina, 2026) (Cowls et al., 2021; Dawson et al., 2023; Sawarkar et al., 2026; Siebold et al., 2017; Ceusters et al., 2022) (Xie et al., 2022; Butler et al., 2012; Cheng et al., 2020b; Berkenkamp et al., 2021; Jujjavarapu & Esfahani, 2022) (Wang et al., 2015; Chang et al., 2025; So et al., 2024; Schafer & Chen, 2020; Breeden & Panagou, 2022) (Li et al., 2010; Liu & McCarthy, 2020; Ren et al., 2025; He et al., 2021; Davenport et al., 2019) (Jung et al., 2011; Marsland et al., 2000; Zheng et al., 2023; Boddington, 2016; Li et al., 2019) (Chen, 2026; Stern & Stiglitz, 2023; Ye et al., 2025; Pasupuleti, 2026b; Taylor et al., 2021) (Tearle et al., 2021; Martinson et al., 2026; Neto, 2006; Ranjan & Kravaris, 2024; Cheng et al., 2020a) (Buerkle et al., 2021; Thananjeyan et al., 2020; Jung et al., 2012; Wang & Long, 2024; Karur et al., 2021) (Liu & Duan, 2021; Paesschesoone et al., 2024; Yang et al., 2021; Lokmic-Tomkins et al., 2023; Kiangala & Wang, 2022) (Yamada et al., 2016; Knuth et al., 2021; Chou et al., 2023; Chemweno et al., 2020; Karagiannis et al., 2022) (Lasota et al., 2017; Cochran et al., 2017; Li et al., 2025; Fan et al., 2024; Wang et al., 2022) (Palleschi et al., 2021; Norouzi et al., 2023; Talamali et al., 2021; Zhao et al., 2025; Tufano et al., 2024) (Tang et al., 2024; Tian et al., 2022; Sun & Lueth, 2023; Kousik et al., 2020; Cao et al., 2025) (Xu et al., 2023; Anderljung et al., 2024; Yang et al., 2022; Franko et al., 2022; Wan et al., 2024) (Burke & Emerick, 2016; Alzubaidi et al., 2023; Sun et al., 2024; Hao et al., 2023; Shao et al., 2021) (Ivanova et al., 2020; Cuan et al., 2025; Matthews, 2025; Juschten et al., 2025; Liu & Li, 2024) (Backer et al., 2022; Didier et al., 2023; Alkadi et al., 2023; Xing et al., 2021; Gagnon et al., 2018) (Liu & Tomizuka, 2014; Zhang et al., 2015; Tian et al., 2020; Xiao, 2024; Robla-Gomez et al., 2017) (Kim et al., 2025; Jabbour & Reddi, 2024; Luo et al., 2024; Larsen et al., 2024; Gualtieri et al., 2020) (Shen et al., 2024; Kanso et al., 2024; Kawaharazuka et al., 2022; Liu et al., 2025b; Miyagawa et al., 2019) (Wang et al., 2023; Ojaghi et al., 2025; Wu et al., 2024; Debus & Himmelrath, 2022; Amaya-Mejia et al., 2022) (Ge et al., 2022; Ducaju et al., 2023; Raiola et al., 2018; Pham et al., 2025; Stoianovici et al., 2016)

B ADDITIONAL THEORY NOTES

The novelty boundary can be stated as a two-level causal claim. A deployment-level safety claim concerns $\tilde{a}_t = F_\phi(x_t, \pi_\theta(x_t))$. A mechanism-level claim concerns $\pi_\theta(x_t)$ before filtering. If the intervention operator changes $\tilde{a}_t$ but leaves $\pi_\theta(x_t)$ unsafe, the result is useful shielding, not evidence that the learned policy has internalized the safety boundary.

The v5 objective attempts to bridge this gap by adding counterfactual unsafe labels and unshielded replay. The audit verifies that these terms matter relative to ablations. The audit also shows they are insufficient under the stronger deployed frontier. That distinction matters: failure against robust MPC is not a proof that the theory is wrong; it is evidence that the present implementation and evidence package are not enough for the claimed venue.

C HOSTILE REVIEWER CHECKLIST

A hostile reviewer would ask whether the method beats action clipping, geometric projection, CBF filters, MPC shields, conformal guards, recovery policies, shielded imitation, safe RL, adversarial safety critics, distillation, and the previous v4 version. This audit answers yes only for a subset of mechanism and calibration questions. It answers no for the decisive deployment and fixed-risk questions. (Sun & Lueth, 2023; Kousik et al., 2020; Cao et al., 2025; Xu et al., 2023; Anderljung et al., 2024; Yang et al., 2022) (Franko et al., 2022; Wan et al., 2024; Burke & Emerick, 2016; Alzubaidi et al., 2023; Sun et al., 2024; Hao et al., 2023) (Shao et al., 2021; Ivanova et al., 2020; Cuan et al., 2025; Matthews, 2025; Juschten et al., 2025; Liu & Li, 2024) (Backer et al., 2022; Didier et al., 2023; Alkadi et al., 2023; Xing et al., 2021; Gagnon et al., 2018; Liu & Tomizuka, 2014) (Zhang et al., 2015; Tian et al., 2020; Xiao, 2024; Robla-Gomez et al., 2017; Kim et al., 2025; Jabbour & Reddi, 2024) (Luo et al., 2024; Larsen et al., 2024; Gualtieri et al., 2020; Shen et al., 2024; Kanso et al., 2024; Kawaharazuka et al., 2022) (Liu et al., 2025b; Miyagawa et al., 2019; Wang et al., 2023; Ojaghi et al., 2025; Wu et al., 2024; Debus & Himmelrath, 2022) (Amaya-Mejia et al., 2022; Ge et al., 2022; Ducaju et al., 2023; Raiola et al., 2018; Pham et al., 2025; Stoianovici et al., 2016)

D RAW FROZEN SUMMARY EXTRACT

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Paper 92 safety_filter_novelty_boundaries v5 expanded audit
Terminal recommendation: KILL_ARCHIVE
ICLR main ready: no
Reason: expanded CPU-only safety-filter audit tests whether filters change the learned policy
mechanism, but standard CBF/MPC/recovery shields still dominate deployed success and safety and
no real robot or accepted high-fidelity safety benchmark evidence exists.
Main rollout rows: 215040
Dataset summary rows: 15360
Main seed-metric rows: 1120
Main metric rows: 1456
Main pairwise rows: 1248
Hard aggregate seed rows: 140
Hard aggregate metric rows: 182
Hard aggregate pairwise rows: 156
Ablation rollout rows: 76800
Ablation seed rows: 100
Ablation metric rows: 130
Stress raw rows: 302400
Stress seed rows: 840
Stress metric rows: 1092
Fixed-risk raw rows: 69120
Fixed-risk seed rows: 480
Fixed-risk metric rows: 288
Fixed-risk pairwise rows: 240
Negative cases: 24

Frozen hard-aggregate gate:
best_success_reference=robust_mpc_shield
safest_reference=robust_mpc_shield
best_unshielded_reference=counterfactual_boundary_learning_filter_v5
lowest_dependence_reference=unfiltered_policy
best_calibration_reference=counterfactual_boundary_learning_filter_v5
best_utility_reference=robust_mpc_shield
best_mechanism_reference=unfiltered_policy
proposal_success=0.22982
best_success=0.47122
proposal_deployed_violation=0.51888
safest_deployed_violation=0.29701
proposal_unshielded_violation=0.89726
best_unshielded_violation=0.89726
proposal_dependence=0.25775
lowest_dependence=0.00000
proposal_boundary_ece=0.10024
best_boundary_ece=0.10024
proposal_utility=-0.24424
best_utility=-0.00351
proposal_mechanism_utility=0.00049
best_mechanism_utility=0.02219
paired_success_lower95=-0.25750
paired_deployed_violation_upper95=0.24007
paired_unshielded_violation_upper95=0.00000
paired_utility_lower95=-0.26567
paired_mechanism_lower95=-0.03102
success_gate=False
deployment_gate=False
mechanism_gate=False
calibration_gate=True
utility_gate=False
ablation_gate=True
mechanism_best_ablation=full_counterfactual_boundary_v5
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stress_gate=False
stress_dominated_by=robust_mpc_shield
fixed_risk_gate=False
scope_gate=False
low_signal_high_risk_shift: v5_coverage=0.00000, v5_success=0.00000,
v5_deployed_violation=0.00000
combined_safety_stress: v5_coverage=0.00000, v5_success=0.00000, v5_deployed_violation=0.00000

Hard aggregate metrics:
unfiltered_policy success=0.03190 deployed=0.93034 unshielded=0.92734 intervention=0.00000
dependence=0.00000 boundary_f1=0.00000 ece=0.63279 transfer=0.93776 utility=-0.61831
mechanism=0.02219
action_clipping_filter success=0.12565 deployed=0.62383 unshielded=0.92617 intervention=0.96094
dependence=0.78200 boundary_f1=0.88653 ece=0.16365 transfer=0.94167 utility=-0.55493
mechanism=-0.29792
geometric_projection_filter success=0.18581 deployed=0.57500 unshielded=0.92852
intervention=0.83125 dependence=0.64565 boundary_f1=0.83036 ece=0.23922 transfer=0.93750
utility=-0.41503 mechanism=-0.23332
cbf_safety_filter success=0.37461 deployed=0.34388 unshielded=0.91875 intervention=0.98828
dependence=0.87853 boundary_f1=0.89589 ece=0.19014 transfer=0.93424 utility=-0.14613
mechanism=-0.31317
robust_mpc_shield success=0.47122 deployed=0.29701 unshielded=0.92018 intervention=0.98997
dependence=0.84285 boundary_f1=0.89764 ece=0.15436 transfer=0.92409 utility=-0.00351
mechanism=-0.29104
conformal_uncertainty_shield success=0.33255 deployed=0.37031 unshielded=0.92005
intervention=0.99141 dependence=0.90956 boundary_f1=0.89766 ece=0.27401 transfer=0.92604
utility=-0.22665 mechanism=-0.33476
recovery_policy_shield success=0.38464 deployed=0.40820 unshielded=0.91641 intervention=0.95755
dependence=0.67078 boundary_f1=0.88648 ece=0.13750 transfer=0.86914 utility=-0.11103
mechanism=-0.20178
shielded_behavior_cloning success=0.12682 deployed=0.71588 unshielded=0.91250
intervention=0.84948 dependence=0.37037 boundary_f1=0.84117 ece=0.17976 transfer=0.89310
utility=-0.49420 mechanism=-0.09936
lagrangian_safe_rl success=0.14805 deployed=0.68620 unshielded=0.90898 intervention=0.82526
dependence=0.32848 boundary_f1=0.82896 ece=0.21393 transfer=0.87695 utility=-0.43902
mechanism=-0.07025
adversarial_safety_critic success=0.13190 deployed=0.67865 unshielded=0.90625
intervention=0.78529 dependence=0.32726 boundary_f1=0.80656 ece=0.27502 transfer=0.86302
utility=-0.45436 mechanism=-0.06553
safety_filter_distillation success=0.19909 deployed=0.58216 unshielded=0.90573
intervention=0.97331 dependence=0.33141 boundary_f1=0.89013 ece=0.13240 transfer=0.84115
utility=-0.32935 mechanism=-0.05857
boundary_learning_filter_v4 success=0.16654 deployed=0.60807 unshielded=0.90221
intervention=0.95300 dependence=0.31556 boundary_f1=0.88242 ece=0.11770 transfer=0.80898
utility=-0.37505 mechanism=-0.04046
counterfactual_boundary_learning_filter_v5 success=0.22982 deployed=0.51888 unshielded=0.89726
intervention=0.97253 dependence=0.25775 boundary_f1=0.89241 ece=0.10024 transfer=0.76693
utility=-0.24424 mechanism=0.00049
oracle_boundary_teacher success=0.43711 deployed=0.21758 unshielded=0.88190 intervention=0.99935
dependence=0.17010 boundary_f1=0.90004 ece=0.02631 transfer=0.64648 utility=0.19902
mechanism=0.11212

Key paired hard-aggregate differences:
counterfactual_boundary_learning_filter_v5_minus_robust_mpc_shield task_success: mean=-0.241406
ci95=0.016092 lower95=-0.257499 upper95=-0.225314
counterfactual_boundary_learning_filter_v5_minus_robust_mpc_shield deployed_violation:
mean=0.221875 ci95=0.018198 lower95=0.203677 upper95=0.240073
counterfactual_boundary_learning_filter_v5_minus_robust_mpc_shield unshielded_violation:
mean=-0.022917 ci95=0.005785 lower95=-0.028702 upper95=-0.017132
counterfactual_boundary_learning_filter_v5_minus_robust_mpc_shield intervention_rate:
mean=-0.017448 ci95=0.004800 lower95=-0.022248 upper95=-0.012648
counterfactual_boundary_learning_filter_v5_minus_robust_mpc_shield intervention_severity:
mean=-0.094490 ci95=0.000971 lower95=-0.095461 upper95=-0.093519
counterfactual_boundary_learning_filter_v5_minus_robust_mpc_shield filter_dependence:
mean=-0.585098 ci95=0.002537 lower95=-0.587636 upper95=-0.582561
counterfactual_boundary_learning_filter_v5_minus_robust_mpc_shield boundary_f1: mean=-0.005233
ci95=0.002562 lower95=-0.007794 upper95=-0.002671
counterfactual_boundary_learning_filter_v5_minus_robust_mpc_shield boundary_ece: mean=-0.054129
ci95=0.008438 lower95=-0.062567 upper95=-0.045690
counterfactual_boundary_learning_filter_v5_minus_robust_mpc_shield transfer_violation:
mean=-0.157162 ci95=0.009123 lower95=-0.166284 upper95=-0.148039
counterfactual_boundary_learning_filter_v5_minus_robust_mpc_shield recovery_success:
mean=-0.236589 ci95=0.020775 lower95=-0.257364 upper95=-0.215813
counterfactual_boundary_learning_filter_v5_minus_robust_mpc_shield robust_utility:
mean=-0.240730 ci95=0.024943 lower95=-0.265673 upper95=-0.215787
counterfactual_boundary_learning_filter_v5_minus_robust_mpc_shield mechanism_utility:
mean=0.291531 ci95=0.005683 lower95=0.285848 upper95=0.297215
counterfactual_boundary_learning_filter_v5_minus_robust_mpc_shield action_regret: mean=0.003261
ci95=0.007931 lower95=-0.004670 upper95=0.011193
counterfactual_boundary_learning_filter_v5_minus_boundary_learning_filter_v4 task_success:
mean=0.063281 ci95=0.008842 lower95=0.054439 upper95=0.072124
counterfactual_boundary_learning_filter_v5_minus_boundary_learning_filter_v4 deployed_violation:
mean=-0.089193 ci95=0.014185 lower95=-0.103378 upper95=-0.075008
counterfactual_boundary_learning_filter_v5_minus_boundary_learning_filter_v4
unshielded_violation: mean=-0.004948 ci95=0.010362 lower95=-0.015310 upper95=0.005413
counterfactual_boundary_learning_filter_v5_minus_boundary_learning_filter_v4 intervention_rate:
mean=0.019531 ci95=0.006731 lower95=0.012800 upper95=0.026262
counterfactual_boundary_learning_filter_v5_minus_boundary_learning_filter_v4
intervention_severity: mean=0.012196 ci95=0.000965 lower95=0.011232 upper95=0.013161
counterfactual_boundary_learning_filter_v5_minus_boundary_learning_filter_v4 filter_dependence:
mean=-0.057807 ci95=0.001995 lower95=-0.059801 upper95=-0.055812
counterfactual_boundary_learning_filter_v5_minus_boundary_learning_filter_v4 boundary_f1:

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mean=0.009984 ci95=0.003833 lower95=0.006151 upper95=0.013817
counterfactual_boundary_learning_filter_v5_minus_boundary_learning_filter_v4 boundary_ece:
mean=-0.017468 ci95=0.006231 lower95=-0.023699 upper95=-0.011237
counterfactual_boundary_learning_filter_v5_minus_boundary_learning_filter_v4 transfer_violation:
mean=-0.042057 ci95=0.009557 lower95=-0.051614 upper95=-0.032500
counterfactual_boundary_learning_filter_v5_minus_boundary_learning_filter_v4 recovery_success:
mean=0.075651 ci95=0.015004 lower95=0.060647 upper95=0.090655
counterfactual_boundary_learning_filter_v5_minus_boundary_learning_filter_v4 robust_utility:
mean=0.130813 ci95=0.016447 lower95=0.114366 upper95=0.147261
counterfactual_boundary_learning_filter_v5_minus_boundary_learning_filter_v4 mechanism_utility:
mean=0.040956 ci95=0.011080 lower95=0.029876 upper95=0.052036
counterfactual_boundary_learning_filter_v5_minus_boundary_learning_filter_v4 action_regret:
mean=-0.053539 ci95=0.005891 lower95=-0.059430 upper95=-0.047649

Ablation utility:
full_counterfactual_boundary_v5 success=0.23320 deployed=0.53385 unshielded=0.89609
dependence=0.25736 utility=-0.25010 mechanism=0.00235
minus_counterfactual_labels success=0.19505 deployed=0.60677 unshielded=0.90951
dependence=0.40916 utility=-0.36732 mechanism=-0.09444
minus_unshielded_replay success=0.19779 deployed=0.59271 unshielded=0.90781 dependence=0.43333
utility=-0.36191 mechanism=-0.10343
minus_intervention_gradient success=0.16693 deployed=0.57747 unshielded=0.90456
dependence=0.37607 utility=-0.38583 mechanism=-0.08589
minus_boundary_margin_loss success=0.18763 deployed=0.60690 unshielded=0.90547
dependence=0.43163 utility=-0.38320 mechanism=-0.10547
minus_recovery_feasibility success=0.15039 deployed=0.60625 unshielded=0.89909
dependence=0.29086 utility=-0.38950 mechanism=-0.03621
minus_calibration_layer success=0.20950 deployed=0.52956 unshielded=0.89583 dependence=0.28032
utility=-0.30660 mechanism=-0.02980
distill_only_boundary success=0.19167 deployed=0.57708 unshielded=0.90299 dependence=0.32696
utility=-0.33246 mechanism=-0.05409
cbf_feedback_only success=0.36601 deployed=0.35352 unshielded=0.93177 dependence=0.87268
utility=-0.15941 mechanism=-0.32390
clipping_feedback_only success=0.12513 deployed=0.62187 unshielded=0.92760 dependence=0.77725
utility=-0.55276 mechanism=-0.29589

Maximum combined stress:
unfiltered_policy success=0.01917 deployed=0.94750 unshielded=0.94444 utility=-0.64228
action_clipping_filter success=0.09750 deployed=0.63861 unshielded=0.95000 utility=-0.59981
geometric_projection_filter success=0.13944 deployed=0.59445 unshielded=0.94750 utility=-0.49038
cbf_safety_filter success=0.31111 deployed=0.36000 unshielded=0.94722 utility=-0.22781
robust_mpc_shield success=0.41528 deployed=0.30417 unshielded=0.94889 utility=-0.07162
conformal_uncertainty_shield success=0.26389 deployed=0.39806 unshielded=0.93944
utility=-0.32284
recovery_policy_shield success=0.32861 deployed=0.41556 unshielded=0.93972 utility=-0.18133
shielded_behavior_cloning success=0.09583 deployed=0.72750 unshielded=0.94167 utility=-0.54259
lagrangian_safe_rl success=0.10944 deployed=0.70639 unshielded=0.93278 utility=-0.49989
adversarial_safety_critic success=0.11472 deployed=0.68250 unshielded=0.93222 utility=-0.48877
safety_filter_distillation success=0.15361 deployed=0.60500 unshielded=0.92972 utility=-0.39508
boundary_learning_filter_v4 success=0.12722 deployed=0.63083 unshielded=0.92528 utility=-0.43591
counterfactual_boundary_learning_filter_v5 success=0.17139 deployed=0.57139 unshielded=0.92639
utility=-0.34202

Fixed-risk budget 0.05:
low_signal_high_risk_shift counterfactual_boundary_learning_filter_v5 coverage=0.00000
accepted_success=0.00000 accepted_deployed_violation=0.00000
accepted_unshielded_violation=0.00000
low_signal_high_risk_shift cbf_safety_filter coverage=0.00000 accepted_success=0.00000
accepted_deployed_violation=0.00000 accepted_unshielded_violation=0.00000
low_signal_high_risk_shift robust_mpc_shield coverage=0.00000 accepted_success=0.00000
accepted_deployed_violation=0.00000 accepted_unshielded_violation=0.00000
low_signal_high_risk_shift conformal_uncertainty_shield coverage=0.00000
accepted_success=0.00000 accepted_deployed_violation=0.00000
accepted_unshielded_violation=0.00000
low_signal_high_risk_shift recovery_policy_shield coverage=0.00000 accepted_success=0.00000
accepted_deployed_violation=0.00000 accepted_unshielded_violation=0.00000
low_signal_high_risk_shift safety_filter_distillation coverage=0.00000 accepted_success=0.00000
accepted_deployed_violation=0.00000 accepted_unshielded_violation=0.00000
combined_safety_stress counterfactual_boundary_learning_filter_v5 coverage=0.00000
accepted_success=0.00000 accepted_deployed_violation=0.00000
accepted_unshielded_violation=0.00000
combined_safety_stress cbf_safety_filter coverage=0.00000 accepted_success=0.00000
accepted_deployed_violation=0.00000 accepted_unshielded_violation=0.00000
combined_safety_stress robust_mpc_shield coverage=0.00000 accepted_success=0.00000
accepted_deployed_violation=0.00000 accepted_unshielded_violation=0.00000
combined_safety_stress conformal_uncertainty_shield coverage=0.00000 accepted_success=0.00000
accepted_deployed_violation=0.00000 accepted_unshielded_violation=0.00000
combined_safety_stress recovery_policy_shield coverage=0.00000 accepted_success=0.00000
accepted_deployed_violation=0.00000 accepted_unshielded_violation=0.00000
combined_safety_stress safety_filter_distillation coverage=0.00000 accepted_success=0.00000
accepted_deployed_violation=0.00000 accepted_unshielded_violation=0.00000

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Negative cases: 24
terminal=KILL_ARCHIVE

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